

ASE 230 Tools - Markdown

Learning Goals

You will be able to:

1. Write a document using basic Markdown syntax.
2. Organize links, images, tables, and code examples.
3. Edit and preview Markdown in VS Code.
4. Convert a Markdown file into a complete HTML document.
5. Check the generated HTML in a web browser

What Is Markdown?

Markdown is a plain-text format for structured documents.

Course Notes

This is ****important****.

- First item
- Second item

The source stays readable as text, and a renderer can display it as a formatted document.

IMPORTANT

Markdown Source and Rendered Output

notes.md

Heading

****bold****

– item

→

→

→

rendered document

large heading

bold text

bulleted list

- You edit the Markdown **source** in a `.md` file.
- VS Code can show a **preview**.

• A converter can create HTML or another output

Create a Markdown File

In VS Code:

1. Select **File → New Text File.**
2. Save the file with the `.md` extension, such as `notes.md`.
3. Confirm that the language indicator shows **Markdown.**
4. Write and save your content.

The `.md` extension tells VS Code and other tools that the file contains Markdown.

Headings

Use one or more  characters followed by a space.

Document Title
Main Section
Subsection
Smaller Subsection

Use one  for the document title, then organize the document in order. Do not choose heading levels only for their visual size.

Paragraphs and Line Breaks

Separate paragraphs with a blank line.

```
This is the first paragraph.
```

```
This is the second paragraph.
```

A normal line break in the source does not always start a new rendered paragraph. Blank lines make the structure explicit and easy to read.

Emphasis

This is **bold** text.
This is *italic* text.
This is ***bold and italic*** text.
Use ``inline code`` for a filename or command.

Rendered meaning:

- **Bold**: strong importance
- *Italic*: emphasis
- `Inline code`: code, commands, variables, or filenames

Lists

Bulleted list:

- PHP
- MySQL
 - Database server
 - Persistent data

Numbered list:

1. Start the server.
2. Open the browser.
3. Check the response.

Indent nested items consistently

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Links

```
[ASE 230 repository](https://github.com/nkuase/ase230)
```

```
[text shown to the reader](destination URL)
```

- Use meaningful link text instead of “click here.”
- Test links in the VS Code preview.
- Include `https://` for an external web address.

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Images

```
![Request and response diagram](pic/request_response.webp)
```

```
![alternative text](image path)
```

- Alternative text explains the image when it cannot be seen.
- A relative path is resolved from the Markdown file's location.
- Upload or copy the image together with the

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Fenced Code Blocks

Use three backticks before and after a block of code.
Add the language name for syntax highlighting.

```
```php
<?php
echo "Hello";
```
```

Do not put the closing backticks on the same line as the code.

Block Quotes and Horizontal Rules

Block quote:

```
> Always validate user input.
```

Horizontal rule:

```
---
```

In Marp slide files used by this course,  also separates slides. Its meaning therefore depends on the document and its position

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Tables

```
Method	Purpose
GET	Read data
POST	Send or create data
```

- The first row contains column headings.
- The second row contains separators.
- Keep a blank line before and after a table.

Tables are widely supported, but they are an

Markdown Is Not Exactly the Same Everywhere

Markdown has multiple implementations and extensions.

- VS Code's built-in preview is based mainly on CommonMark.
- GitHub supports GitHub Flavored Markdown features such as tables and task lists.
- Pandoc supports its own extended Markdown syntax

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Use Markdown in VS Code

VS Code supports Markdown without installing an extension.

1. Open a `.md` file.
2. Edit the Markdown source normally.
3. Save with `Ctrl+S` on Windows/Linux or `Cmd+S` on macOS.
4. Open the preview:
 - Windows/Linux: `Ctrl+Shift+V`

Preview Beside the Source

To edit and preview at the same time:

- Windows/Linux: `Ctrl+K`, then `V`
- macOS: `Cmd+K`, then `V`

You can also:

1. Open the Command Palette.
2. Search for **Markdown: Open Preview to the Side**.
3. Select the command.

Preview Is Not an HTML File

VS Code's built-in Markdown preview displays rendered content inside VS Code.

It does **not** automatically create a reusable `.html` file.

| Difference | VS Code Preview | HTML Conversion |
|------------------|-----------------|---------------------------------------|
| Where it appears | Inside VS Code | In a separate <code>.html</code> file |
| | | |

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Convert Markdown to HTML with Pandoc

Pandoc is a document conversion program.

```
notes.md → Pandoc → notes.html
```

Pandoc is separate from VS Code. You write the file in VS Code and run Pandoc in the integrated terminal.

Official website: <https://pandoc.org>

Install Pandoc

Ubuntu / WSL2:

```
sudo apt update  
sudo apt install -y pandoc
```

macOS with Homebrew:

```
brew install pandoc
```

Verify the installation:

```
pandoc --version
```

Open the VS Code Terminal

Open the folder containing your Markdown file in VS Code, then open the integrated terminal:

Terminal → New Terminal

Shortcut:

- Windows/Linux: **Ctrl+`**
- macOS: **Cmd+`**

Check that the terminal is in the correct folder before converting the file

Create a Complete HTML Document

If the Markdown file is `notes.md`, run:

```
pandoc notes.md --standalone --output notes.html
```

Short form:

```
pandoc notes.md -s -o notes.html
```

- `notes.md`: input file
- `-s` / `--standalone`: include a complete

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Why `--standalone` Matters

Without `--standalone`, Pandoc normally creates an HTML fragment:

```
<h1>Course Notes</h1>  
<p>Hello.</p>
```

With `--standalone`, it creates a complete document containing elements such as:

```
<!DOCTYPE html>  
<html>
```

Add a Page Title

Pandoc can read document metadata from YAML at the top of the Markdown file:

```
---  
title: ASE 230 Notes  
author: Your Name  
lang: en  
---  
  
# Introduction
```

Then convert it normally:

Add CSS Styling

Create a file named `style.css`:

```
body {  
  max-width: 800px;  
  margin: 2rem auto;  
  font-family: Arial, sans-serif;  
  line-height: 1.6;  
}  
code { background: #f2f2f2; }
```

Then link it during conversion:

```
pandoc notes.md -s --css style.css -o notes.html
```

Create One Self-Contained HTML File

If the Markdown uses local images or CSS, Pandoc can embed those resources into the HTML:

```
pandoc notes.md -s \  
  --css style.css \  
  --embed-resources \  
  -o notes.html
```

The resulting HTML is easier to move or submit because its local CSS and images are embedded in

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Open and Check the HTML

After conversion:

1. Find `notes.html` in the VS Code Explorer.
2. Reveal it in Finder or File Explorer.
3. Double-click it to open it in a web browser.
4. Check headings, lists, links, images, tables, and code blocks.

If an image is missing, check its relative path and confirm that the image was copied with the HTML

Optional: Use a VS Code Extension

VS Code extensions can add HTML export commands.

1. Open Extensions: `Ctrl+Shift+X` / `Cmd+Shift+X`.
2. Search for a Markdown extension that supports HTML export.
3. Read its publisher, permissions, recent updates, and documentation

Common Problems

Problem	What to check
Preview shows plain text	Confirm the filename ends in <code>.md</code> .
<code>pandoc: command not found</code>	Install Pandoc and reopen the terminal.
Wrong file converted	Check the terminal folder and input filename.
HTML is only a fragment	Add <code>-s</code> or <code>--standalone</code> .

Final Checkpoint

You should now be able to:

1. Create a `.md` file and use common Markdown syntax.
2. Open VS Code's built-in Markdown preview.
3. Explain the difference between previewing and converting.
4. Convert Markdown into standalone HTML with Pandoc.
5. Add metadata and CSS to the generated HTML